



Supplementary Artifact Audit

Measuring Compact Local Evidence Carriers

A Reproducibility Audit of Qorx

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This supplement provides the evidence inventory, receipt-level derivations, claim register, and confirmatory protocols behind the accompanying research report. It separates the measured May path, the public Qorx 1.0.6 fixtures, and the later development snapshot. Readers can use the inventory to verify each reported quantity, identify the current reproducibility level, and see exactly what a source-to-binary or environmental study would add.

A Evidence inventory and provenance

This appendix organizes the evidence into four reproducibility levels: integrity-verifiable receipts, protocol-level repeatability, same-author replication, and independent source-to-binary reproduction. The May directory supports the first two levels and documents a run labeled cross-droplet reproduction. Reaching the fourth level requires the exact source revision, dependency environment, compiler, and build attestation.

A.1 Inspected evidence set

The primary evidence directory used for this manuscript is identified in the accompanying deposit candidate as:

```
evidence/qorx-amd-x1-results-2026-05-30/
```

The directory contains three compressed run bundles, raw JSON, logs, two reproduction scripts, claim-boundary notes, and digest files. The reconstruction scripts recover the public corpus slices, while the exact measured source tree remains the additional input needed for a source-to-binary rebuild. The following inventory names every item used in the audit.

Item	Evidence used in this revision
Primary archive	<code>qorx-context-reduction-x1-fever12500k-20260529T163955Z.tar.gz</code>
Companion archives	<code>qorx-context-reduction-x1-fever10m-fix2-20260529T162853Z.tar.gz</code> ; <code>qorx-context-reduction-x1-repro-nofever-20260529T162159Z.tar.gz</code>
Raw measurement	<code>qcr/qorx-context-reduction-x1-fever12500k-20260529T163955Z.json</code> inside the archive
Method note	<code>QORX-CONTEXT-REDUCTION-METHOD-REFERENCE.txt</code>
Final report	<code>FINAL-X1-QORX-CONTEXT-REDUCTION-12500K-REPORT.txt</code>
Reproduction declaration	<code>repro-qorx-context-reduction-fever12500k.sh</code>
Companion rerun script	<code>repro-qorx-context-reduction-fever10m.sh</code>
Author-held corroboration	<code>docs/benchmarks/qorx-context-reduction-x1-fever12500k-2026-05-30.json</code> ; not used as the primary receipt
Public implementation check	https://github.com/bbbrainfuckk/qorx/blob/fdf7589baf2fa16a9129307658a24c4c7a874145/docs/benchmarks/live.json
Provider comparison	https://github.com/bbbrainfuckk/qorx/blob/362e142f5984147fafdab365a4f24e203fc0ac7a/docs/benchmarks/datacamp-provider-comparison-2026-07-22.json

Table 1: Evidence layers inspected for the report. The July artifacts test Qorx 1.0.6 as a separate evaluation layer from the May binary.

A.2 Digest audit

The headline archive digest was recomputed directly from the compressed file on July 22, 2026. It matched the value recorded in the receipt:

```
archive SHA-256:
09dd67e3e15913cb1e71f575f525014a3184e73ebf58e767a400fc933d7c2332

recorded Qorx binary SHA-256:
42be06b3991c646aa42eed8bf17b76ba2afa067638123309fa41a599ebfa15c

recorded harness SHA-256:
6e6a64557c4ab34962a5d144ba6efa2aea2120f09ad2e265e771fc2ad19f1d23
```

On July 23, 2026, all three outer sidecar digests were rechecked and all thirteen entries in each archive's internal SHA256SUMS.txt were recomputed without mismatch. The companion archive SHA-256 values are 9d59b9df024f296726a2bcc605dc41ce30f7bb607015f303fbc5207815f20358 for the FEVER 10M run and f9d87b602b0cc4ad636db8d44a7790572f3cc981129b0f5f9c175f502a6dd8ce for the no-FEVER run labeled cross-droplet reproduction.

A digest supports identity and tamper detection for the bytes that were hashed. Source identity, build provenance, and workload generalization require additional links: a source-to-binary provenance chain, a reproducible build or equivalent attestation, and a broader task design.

A.3 Evidence hierarchy

Evidence level	What it establishes	Next evidence needed
Raw JSON row	Recorded query, handle count, carrier bytes, estimator output, and timing field	Source correctness, retrieval quality, downstream answer quality
Archive digest	Byte identity of the surviving receipt	Relationship between source, binary, corpus, and receipt
Method note	Declared carrier layout and estimator	Independent confirmation that every runtime path followed the declaration
Rerun scripts	Public-dataset download, splitting, corpus assembly, benchmark invocation, and gate logic	Exact source tree, prebuilt executable, compiler, and dependency identity
Companion May archives	Same recorded binary and harness hashes, successful gates at three corpus sizes, and a documented cross-droplet label	Independent host attestation and source-to-binary reconstruction
Author-held copy	Corroborating copy of selected receipt fields	Public inspection and independent rebuilding
Public 1.0.6 source	Inspectable current implementation and scripts	The historical May 2026 measured implementation
Provider receipt	Provider-reported token and fixture fields for three July tasks	Repeated trials, matched cache policy, environmental telemetry

Table 2: Current contribution and next verification step for each evidence layer.

B Complete May receipt observations and derivations

The archived JSON contains exactly five query observations. No row was removed as warm-up, and the descriptive statistics in the paper include the visibly slower first observation.

ID	Query	Handles	Bytes	Est. tokens	Core ms
Q1	qorx codex hooks quark pointer mcp	2	56	14	3.512
Q2	context fault proof pages resolver boundary	2	56	14	0.396
Q3	strict answer refusal unsupported claims	2	56	14	0.208
Q4	qlp quantum leap qubo local quarks	2	56	14	0.242
Q5	benchmark reproduction external artifact boundaries	2	56	14	0.129

Table 3: All raw query rows in the May receipt.

The method note defines the carrier as an eight-byte header followed by 24 bytes for each selected quark handle. With two handles, the declared byte count is

$$8 + 2(24) = 56 \text{ bytes.} \quad (1)$$

The artifact uses a deterministic `char4` estimate rather than a named provider tokenizer:

$$\tau_{\text{char4}}(x) = \left\lceil \frac{|x|_{\text{bytes}}}{4} \right\rceil, \quad \tau_{\text{char4}}(H_q) = \left\lceil \frac{56}{4} \right\rceil = 14. \quad (2)$$

The recorded corpus estimate is 184,789,445 tokens under the receipt’s estimator. The reported quotient is therefore

$$\frac{184,789,445}{14} = 13,199,246.07142857 \dots \quad (3)$$

This quotient is an index-to-handle addressing ratio. It is not an entropy ratio, a provider-billing ratio, or evidence that 184,789,445 model-readable tokens were losslessly represented by fourteen tokens. Shannon’s information theory distinguishes code length from the information properties of a source and channel [6]; the receipt contains no entropy model or semantic reconstruction test.

B.1 Latency recomputation and boundary

Sorting the recorded core times gives 0.129, 0.208, 0.242, 0.396, and 3.512 ms. Their arithmetic mean is

$$\bar{t} = \frac{3.512 + 0.396 + 0.208 + 0.242 + 0.129}{5} = 0.8974 \text{ ms}, \quad (4)$$

and the middle order statistic is 0.242 ms. These values reproduce the receipt exactly. The small sample does not support a distributional model, confidence interval, or tail-latency claim.

Field	Value	Unit	Interpretation
avg_latency_ms	0.8974	ms	Mean of the five in-process lookup observations
max_latency_ms	3.512	ms	Maximum of the five observations
quas_build_latency_ms	12,239.854	ms	Separate build field recorded by the harness
quas_wall_latency_ms	18,049.631	ms	Separate wall field recorded by the harness
elapsed_seconds	175.325	s	Full recorded suite duration

Table 4: Distinct timing fields retained rather than conflated.

The core result excludes command-line startup and index loading. It is therefore not an end-to-end latency result. A future report should record cold start, warm start, index mapping, query processing, evidence resolution, serialization, transport, provider inference, and total task completion as separate spans.

B.2 Missing outcome fields

The May JSON does not store resolved evidence excerpts, a relevance judgment, retrieval recall, an answer, a task oracle, provider token counts, provider cost, a power trace, or facility telemetry. It also fixes the carrier at two handles, so it cannot show how carrier size changes with query complexity. The absence of those fields is a limitation of the recorded experiment, not a negative result about the unmeasured components.

C Public Qorx 1.0.6 repository check

The July 22 repository check is useful because it exercises public source and emits a normalized machine-readable report. It is not joined statistically with the May receipt. The run identifies Qorx 1.0.6 at commit `fdf7589`, indexes 748 quarks, and records 407,902 local `char4` tokens. The principal commands and outputs are summarized below.

Operation	Elapsed ms	Visible/used	Omitted	Recorded boundary
Index	389.895	748 quarks	—	407,902 estimated local tokens
Session	95.380	69 tokens	407,833	Local pointer; ratio 5,911.62x
Bench, query 1	—	410 tokens	407,492	One selected quark; ratio 994.88x
Bench, query 2	—	448 tokens	407,454	Two selected quarks; ratio 910.50x
Bench, query 3	—	476 tokens	407,426	Three selected quarks; ratio 856.94x
Squeeze	235.451	448 tokens	407,454	Extractive, local, zero provider calls
Agent	429.408	411 packed tokens	407,491	Local planner; zero provider calls

Table 5: Selected fields from the public repository check. Elapsed command times are not compared with the May core lookup field.

The session pointer is 69 estimated tokens while the local index remains on the host. Its 5,911.62x value is a local accounting quotient. The pack and squeeze paths expose larger evidence payloads. This distinction illustrates why carrier size and resolved evidence size require separate fields.

C.1 Strict supported and refusal fixtures

Two deterministic strict-answer fixtures were recorded:

Question	Expected	Observed	Pass	Used	Evidence rows
Context injection is a compact agent contract	supported	supported	yes	325	3
galactic banana escrow treaty	not found	not found	yes	8	0

Table 6: Repository fixtures. The second question is deliberately absent from the indexed source.

The supported fixture checks exact indexed strings; the refusal fixture checks that an absent phrase returns no extractive answer. Passing one of each demonstrates the disclosed control path. It does not estimate false-support or false-refusal rates over a population of realistic tasks.

C.2 Evidence identity within the public check

The squeezed evidence retained path, line range, token count, matched terms, and excerpt hash. Those fields are useful for traceability because a reviewer can locate the exact source region. They are not cryptographic proof of semantic correctness. Proof-carrying code requires an explicit safety policy, a formal proof, and a checker [3]; Qorx’s July evidence records are better described as citation- and hash-carrying excerpts unless and until a formal proof system is specified.

D DataCamp provider comparison: full disclosure

The provider comparison used the Qorx 1.0.6 static Linux release in separate DataCamp OpenAI and Anthropic sandboxes. The synthetic corpus contained 106,758 characters. Its identities were:

```
release archive SHA-256:
c22dd19e666d7a12b19f80c9ff78922fe22abe6d423c45a3f32befe41d1c41ef
binary SHA-256:
1cbee5289a75610cbf8c812f4ce5626435712f1aae2e684a5d90565a7ac728c2
synthetic corpus SHA-256:
517c10fa62eb818082c58d6e3de879d6ca2962eeb4369476f1c368df22d35288
```

The system instruction required answers only from supplied context and the exact string NOT_SUPPORTED when every requested fact was not present. Temperature was zero, the output cap was 80 tokens, and each question was run once in the full condition followed by once in the Qorx condition.

Task	Question	Deterministic expectation
Supported operations	What is the emergency production override review window, and who must approve it?	“27 minutes” and “Reliability Lead”
Supported release	What is the Atlas release marker, and who owns the release?	“cobalt-heron-73” and “Mina Sol”
Unsupported customer	What is the customer’s bank account number?	Exact NOT_SUPPORTED

Table 7: All three synthetic provider-comparison tasks.

Provider	Task	Full input	Qorx input	Qorx pack ms	Full elapsed ms	Qorx elapsed ms
OpenAI	operations	21,953	336	11.801	2,892.554	1,041.064
OpenAI	release	21,951	332	14.053	1,200.635	658.205
OpenAI	unsupported	21,946	146	11.822	660.160	590.461
Anthropic	operations	25,822	402	12.963	2,700.457	1,372.947
Anthropic	release	25,820	398	14.274	3,090.262	3,280.373
Anthropic	unsupported	25,816	163	14.025	5,051.017	908.977

Table 8: Every provider row in the July comparison. Times are sequential observations, not controlled latency estimates.

OpenAI reported 65,850 aggregate full-condition input tokens and 814 Qorx-condition tokens, giving 80.8968x. Anthropic reported 77,458 and 963, giving 80.4341x. All twelve task-condition results met their deterministic expectations: three of three tasks for each condition and adapter. Two OpenAI full-context calls reported 21,888 cached input tokens; the Anthropic adapter reported none. That cache asymmetry and the fixed request order invalidate a controlled latency comparison. Table 8 is retained for transparency, not as a speed claim.

The July fixture improves on the May receipt by recording model-facing evidence and answers. It remains preliminary because it contains three synthetic questions, one run per cell, fixed order, heterogeneous provider caching, and substring-based scoring. A future comparison should randomize order, run enough repetitions for interval estimation, include adversarial near-miss questions, and use blinded task-level judgments or executable oracles.

E Post-freeze source inspection

This inspection covers an author-supplied local source snapshot dated July 31, 2026. It is separate from the May receipts and the July 22 public fixtures. The snapshot identifies its Cargo package as `qorx 0.0.1-a-ylem`, but its containing repository has no commit. A deterministic manifest over 319 source and configuration files, excluding build products and binaries, produced SHA-256 `8f7c66a6...4fc240f0`. The compiler was `rustc 1.95.0` and the package manager was `cargo 1.95.0`.

The compiled qorx test target reported 363 passed tests and no failures. `cargo check --all-targets` completed successfully with warnings. `cargo fmt --check` failed on `src/bin/idra-source-worker.rs`. Four feature files present in the source tree, `auto_index.rs`, `index_store.rs`, `qvp.rs`, and `zero.rs`, were not declared by the crate root and were therefore absent from the compiled target. Passing tests do not cover those files.

Binary provenance is also unresolved. The installed `qorx.exe` digest was `7e9a5347...2f37b2f`, while the located distribution build digest was `f6ac08ad...f63ce33`. A version string match does not establish source identity when executable hashes differ and no attested build record links them.

The IDRA pipeline uses structured argument arrays, output redaction, changed-file checks, and post-run secret scanning. Its worker adapters also use permission-bypass flags, inherit most host environment variables, and rely on instructions rather than an operating-system sandbox to prevent protected-file reads. The snapshot therefore does not support a claim that a compromised or misbehaving worker cannot read host secrets. This is a static boundary finding, not an observed breach.

F Claim-to-evidence register

This register is the controlling interpretation of the report. A claim is marked *verified* when the relevant arithmetic or field was independently checked; *fixture-supported* when it holds only for a disclosed deterministic task; *development status* when it describes inspected code without an outcome evaluation; and *unmeasured* when no suitable observation exists.

Table 9: Claim-to-evidence register.

Claim	Status	Evidence boundary
May archive digest matches the recorded digest	Verified	Recomputed from the surviving compressed receipt
All three May outer and internal digests verify	Verified	Sidecars match; thirteen internal members per archive recomputed without mismatch
May rerun procedure is available	Verified artifact	Two scripts reconstruct public dataset slices and invoke the benchmark when the required source tree and executable exist
No-FEVER run is a cross-droplet reproduction	Documented label	Companion README states cross-droplet; archive does not independently encode host identity
Exact May source-to-binary rebuild is available	Not established	Recorded binary and harness hashes exist, but their exact source revision and attested build environment are absent
May indexed estimate is 184,789,445 tokens	Verified field	Receipt's deterministic estimator; not a provider tokenizer
Two-handle carrier is 56 bytes	Verified derivation	Eight-byte header plus two 24-byte handles

Claim	Status	Evidence boundary
Carrier estimate is fourteen tokens	Verified derivation	<code>char4</code> estimator only
Addressing ratio is 13,199,246.07x	Verified derivation	Index estimate divided by handle estimate; not semantic compression
Mean core lookup is 0.8974 ms	Verified descriptive statistic	Five in-process rows; excludes startup and index loading
Median core lookup is 0.242 ms	Verified descriptive statistic	Middle of five sorted values
Explicit May provider-call count	Not recorded	The method identifies a local in-process path, but the JSON has no provider-call counter
May retrieval is accurate or sufficient	Unmeasured	No resolved excerpts or relevance labels
May answers are correct	Unmeasured	No downstream answers or task oracle
Public session pointer ratio is 5,911.62x	Fixture-supported	407,902 local estimated tokens divided by 69 visible estimated tokens
Public strict supported task passed	Fixture-supported	One exact repository phrase
Public strict refusal task passed	Fixture-supported	One deliberately absent phrase
OpenAI provider-input ratio is 80.90x	Fixture-supported	Three synthetic tasks, one call per condition
Anthropic provider-input ratio is 80.43x	Fixture-supported	Three synthetic tasks, one call per condition
Provider latency is reduced	Unmeasured	Fixed order and heterogeneous caching prevent causal inference
Provider cost is reduced	Unmeasured	No normalized price/cost boundary in the receipt
Qorx reduces hallucinations generally	Unmeasured	Three fixture passes cannot estimate hallucination rate
Qorx protects private data	Unmeasured	No threat model or leakage experiment in the receipt
Qorx reduces energy, carbon, or water	Unmeasured	No paired telemetry or facility boundary
Post-freeze compiled Qorx tests pass	Verified local run	363 compiled tests passed; this is not an independent reproduction
Post-freeze draft feature files are compiled	Not established	Four named files are outside the crate module graph
Post-freeze installed binary matches inspected source	Not established	No commit or attested build record; located executable digests differ
IDRA workers are isolated from host secrets	Not established	Policy and post-run checks are present, but no operating-system sandbox enforces the read boundary

The register should travel with any summary of the paper. If later work supplies stronger evidence, a new revision should update the status and cite a versioned receipt rather than silently broadening the interpretation of the May artifact.

G Preregistered environmental evaluation protocol

The main paper states a mechanism and accounting equations. This appendix turns that mechanism into a decision-complete preregistration template. It is intentionally stricter than the current evidence so that a future result cannot be selected after inspecting favorable outcomes.

G.1 Research questions and hypotheses

For a fixed task distribution, declared quality gate, model, hardware class, cache policy, and region, compare a baseline condition B that sends the declared full context with a Qorx condition Q that builds or reuses a local index and sends resolved evidence.

Primary energy hypothesis. The mean end-to-end operational energy per successful task is lower under Q after amortizing index construction over the observed reuse count.

Quality non-inferiority hypothesis. Task success and supported-refusal quality under Q are no worse than a preregistered margin relative to B .

Carbon hypothesis. Location- and time-matched operational plus allocated embodied carbon per successful task is lower under Q .

Water hypothesis. Water consumption and withdrawal per successful task are reported separately and are lower under Q within a declared electricity and facility boundary.

Failure to pass the quality gate makes an efficiency comparison invalid: an incomplete or incorrect task is not the same functional unit as a correct task.

G.2 Design and randomization

Use at least three repositories not used to author the queries. Freeze the task set before running either condition. For each task, randomize the order of B and Q ; where provider caching can cross-contaminate conditions, isolate accounts, cache namespaces, or sufficiently separated prefixes. Record cold and warm trials separately. Preregister the number of runs, exclusion rules, retry policy, timeout, and analysis model. Publish failed and timed-out runs.

The unit of randomization should match the expected interference boundary. If multiple tasks share an index or cache, randomize at the repository-session block rather than pretending that task rows are independent. Use paired analysis when both conditions operate on the same task and comparable system state.

G.3 Measurement boundary

The minimum span inventory is:

1. corpus scan, parsing, embedding or sparse-index construction, serialization, and storage writes;
2. index load or memory mapping, local retrieval, selection, evidence extraction, and carrier serialization;
3. request construction, network transfer, provider prefill, decoding, and tool calls;
4. retries, refusals, failed attempts, validation, and post-processing;
5. idle or reserved capacity allocated to the workload; and
6. embodied-hardware allocation when material to the declared Software Carbon Intensity boundary [1].

Meter local CPU, accelerator, memory, and storage energy at a sampling rate appropriate to the task duration. Record device power and host power separately when possible. Provider-side energy should come from provider telemetry or an explicitly labeled model; local measurements must not be presented as end-to-end energy when provider inference is outside the meter.

G.4 Amortization and break-even

Let E_{index} be one-time index energy, E_{resolve} per-task local resolution energy, and $E_{\text{remote},Q}$ the transport plus provider energy under Q orx. With N observed reuses,

$$E_Q(N) = \frac{E_{\text{index}}}{N} + E_{\text{resolve}} + E_{\text{remote},Q}. \quad (5)$$

Report the whole curve over plausible N , not only a favorable reuse count. If the stationary saving $E_B - (E_{\text{resolve}} + E_{\text{remote},Q})$ is non-positive, no reuse count amortizes indexing under that configuration. If it is positive, the estimated break-even count is

$$N_{\text{break-even}} = \left\lceil \frac{E_{\text{index}}}{E_B - (E_{\text{resolve}} + E_{\text{remote},Q})} \right\rceil.$$

G.5 Carbon, water, and uncertainty

For every energy interval, record location, time, electricity source or marginal/average factor choice, and factor provenance. Report operational energy before multiplying by a carbon factor. Keep embodied allocation visible. For water, distinguish direct facility consumption, facility withdrawal, and electricity-related water; do not combine them into an unlabeled single number. Li et al. show why workload location and water boundary matter [2], while the LBNL report supplies data-centre system context rather than a Qorx-specific factor [7].

Propagate meter error, factor uncertainty, run-to-run variance, and missing provider telemetry. Publish a sensitivity table across cache policies, reuse counts, grid factors, and water factors. A result is inconclusive when the uncertainty interval crosses zero or when quality non-inferiority fails.

G.6 Minimum result table

Outcome	Required statistic	Required boundary note
Task quality	Success, refusal precision/recall, retries	Oracle, margin, adjudication procedure
Tokens and bytes	Input, output, cached input, evidence bytes	Estimator/tokenizer and cache semantics
Latency	Median and interval by span	Cold/warm state and request order
Energy	Joules per successful task and index joules	Meter, sampling, provider inclusion
Carbon	gCO ₂ e per successful task	Location, time, operational/embodied split
Water consumption	Litres per successful task	Direct/electricity boundary and source
Water withdrawal	Litres per successful task	Kept separate from consumption
Break-even	Reuse-count curve	Index lifetime and stationary-term assumption

Table 10: Minimum environmental result disclosure.

H Retrieval, answer quality, and continuity evaluation

A useful context system must be evaluated on more than carrier size. The principal causal chain is: query selects evidence; evidence supports a decision; the decision advances a task; continuity preserves the relevant state across interruption. Each link needs an observable outcome.

H.1 Baselines

At minimum, compare:

1. full transcript or declared full-context baseline;
2. lexical retrieval such as BM25;
3. a dense retrieval baseline using a frozen model;
4. a hybrid lexical–dense condition where practical; and
5. Qorx with carrier bytes and resolved evidence bytes reported separately.

All conditions must receive the same task statement, output budget, tool permissions, model revision, retry budget, and oracle. If a baseline cannot fit within a provider context window, report truncation behavior rather than silently dropping the task.

H.2 Retrieval and sufficiency metrics

For a query with labeled relevant evidence set G_q and retrieved set R_q , report recall and precision:

$$\text{Recall}(q) = \frac{|R_q \cap G_q|}{|G_q|}, \quad \text{Precision}(q) = \frac{|R_q \cap G_q|}{|R_q|}. \quad (6)$$

Also record evidence bytes, estimated and provider tokens, unique source files, duplicated lines, citation validity, and whether the retrieved set is sufficient to answer the task. RAGChecker and SURE-RAG motivate separating retrieval and answer-level sufficiency rather than collapsing them into one score [5, 4].

H.3 Answer and refusal metrics

Supported tasks require executable tests, exact structured answers, or blinded expert judgments defined before inference. Unsupported and partially supported tasks require an abstention oracle. Report:

- task success and first-correct-attempt rate;
- unsupported-claim rate per answer and per atomic claim;
- refusal precision, refusal recall, and selective risk by confidence threshold;

- citation precision and the fraction of answer claims entailed by cited lines;
- number of retries, tool calls, and tokens until first correct action; and
- failure categories, including missing evidence, wrong retrieval, wrong synthesis, stale state, and invalid refusal.

The phrase “zero hallucination” should be reserved for a declared finite test set in which the adjudication protocol found zero unsupported claims, accompanied by an uncertainty interval. It should not describe a system generally.

H.4 Continuity protocol

Long-horizon continuity requires interruption, not merely a long-running process. Construct tasks that produce files, decisions, unresolved questions, and tool state. At preregistered checkpoints, terminate the process, remove transient conversation context, and resume from the declared local state. Score whether the system:

1. identifies the correct project and last completed action;
2. retrieves the exact artifacts and evidence needed for the next decision;
3. avoids repeating completed work;
4. preserves unresolved risks and user constraints;
5. refuses when the stored state cannot prove a requested fact; and
6. completes the task under the same oracle as an uninterrupted baseline.

Report interruption interval, number of compactions, number of resumptions, state size, stale-state incidents, recovery time, and final task success. AMA-Bench provides a relevant long-horizon evaluation direction [8]; no such result is present in the inspected Qorx artifacts.

I Privacy, security, and provenance boundaries

Local indexing changes where data resides and when it is transmitted, but locality alone is not a security proof. A defensible evaluation begins with assets, adversaries, trust boundaries, and observable leakage.

I.1 Threat model template

Element	Minimum declaration	Example measurement
Protected assets	Source, credentials, client records, privileged logs, index metadata	Seeded secrets and classified source regions
Adversaries	Remote provider, malicious local user, compromised plugin, network observer	Controlled exfiltration and unauthorized-resolution attempts
Trusted computing base	Qorx binary, OS, filesystem permissions, proxy, dependencies	Versioned manifest and privilege audit
Data at rest	Index, cache, logs, temporary files, crash dumps	File inventory, permissions, encryption state, secure-deletion behavior
Data in transit	Carrier, resolved evidence, request and response bodies	Packet or proxy trace with byte classification
Authorization	Who may resolve which handle for which workspace	Cross-user and cross-project access tests
Integrity	Source-to-excerpt mapping, digest validation, stale index handling	Tamper, replay, substitution, and rollback tests

Table 11: Minimum threat-model disclosure for a local context resolver.

The May receipt identifies the measured path as local and in-process but does not contain a provider-call counter. That declared boundary does not show that other processes could not read the index, that a carrier reveals no metadata, that later resolution transmits no sensitive text, or that a compromised local agent cannot request broad evidence.

I.2 Leakage experiment

Create a corpus with labeled public, confidential, secret, and credential-like spans. Define authorized and unauthorized tasks and users. Record whether each condition exposes raw bytes, identifiers, filenames, line ranges, hashes, or statistical metadata from restricted spans. Test direct queries, paraphrases, wildcard-like queries, repeated narrowing, cross-workspace handles, stale handles, copied carriers, log inspection, and malformed resolver requests. A successful result needs both low unauthorized disclosure and high authorized retrieval; simply refusing every request is not useful privacy.

I.3 Provenance language

Hashes in Qorx receipts identify bytes and make later mutation detectable. They do not by themselves establish authorship, build provenance, semantic truth, or temporal ordering. A stronger provenance story would publish signed manifests that bind source revision, dependency lockfile, compiler, build command, corpus digest, binary digest, runtime configuration, and result digest. Reproducible builds or attestations would then let an independent party test the binding.

The term “proof” in a product field must be qualified. A citation, digest, or receipt is useful evidence; it becomes a formal proof only when the proposition, logic, assumptions, proof object, and verification algorithm are specified. This distinction keeps Qorx’s evidence-first objective compatible with the established proof-carrying-code lineage [3] without claiming an implementation that the artifacts do not show.

J Reproduction package and reporting schema

A public revision should deposit an immutable package rather than relying on a moving repository branch. The recommended layout is:

```
qorx-study-<date>/
  README-FIRST.md
  LICENSE
  CITATION.cff
  manifest.sha256
  source/
  build/
    toolchain.lock
    build-command.txt
  corpus/
    manifest.json
  tasks/
    tasks.jsonl
    annotations.jsonl
  runs/
    raw-events.jsonl
    normalized-results.json
    failures.jsonl
  environment/
    hardware.json
    software.json
    provider.json
  analysis/
    reproduce.sh
    tables-and-figures.ipynb
  paper/
    manuscript.tex
    supplement.tex
```

The manifest should bind every byte needed to interpret the result. Large or restricted corpora may require a content-addressed manifest and access procedure instead of direct redistribution, but the paper must still state exactly what an independent reviewer cannot obtain.

J.1 Minimum normalized event fields

Table 12: Minimum fields for a future normalized Qorx evaluation event.

Field group	Example fields	Purpose
Identity	study, run, task, condition, repository, source commit	Join records without depending on filenames
Software	Qorx version, binary digest, compiler, dependency lock digest	Establish executable identity and rebuild boundary
Corpus	corpus digest, file count, bytes, estimator, estimated tokens	Declare what was indexed and how size was estimated
Carrier	scheme, bytes, estimator, tokens, handle count	Measure the address separately from evidence
Evidence	source bytes/tokens, resolved bytes/tokens, paths, line ranges, excerpt digests	Measure payload and traceability
Retrieval	rank, score, relevant labels, sufficiency label	Calculate recall, precision, and evidence sufficiency
Task outcome	oracle, success, refusal class, unsupported claims, citations	Tie efficiency to functional quality
Timing	cold/warm, index, load, query, resolve, transport, provider, total	Prevent incompatible spans from being conflated
Provider	provider, model revision, input/output/cached tokens, region, request order	Interpret model-facing counts and cache effects
Energy	device, meter, sampling interval, joules by span	Support measured rather than assumed energy claims
Impact factors	carbon source, water source, location, time, uncertainty	Make carbon and water arithmetic reproducible
Failures	timeout, retry, exception, exclusion reason	Prevent successful-only reporting

J.2 Author and independent-review checklist

Before deposit, the author should be able to answer yes to each item:

1. Does every quantitative claim point to a versioned raw field or an explicit derivation?
2. Are carrier size, resolved evidence size, model input, and indexed-corpus estimate separate?
3. Are local, end-to-end, and provider-side latency spans labeled distinctly?
4. Are all task exclusions, failures, retries, and cache states published?
5. Is the measured source revision public or is its absence stated prominently?
6. Can an independent reviewer rebuild the binary or verify why that is impossible?
7. Are quality and refusal thresholds fixed before comparing efficiency?
8. Are energy, carbon, and water presented only when the corresponding boundary was measured?
9. Are development targets separated from observed endurance or concurrency?
10. Are physics-inspired terms labeled as vocabulary unless a physical or mathematical model is actually evaluated?
11. Are conflicts of interest, language-model assistance, licensing, and author responsibility disclosed?
12. Does the final public PDF match the source and digest manifest in the deposit?

This manuscript answers the first, second, third, eighth, ninth, tenth, and eleventh items and documents integrity-checked protocol repeatability across three May archives. The next reproducibility step is an independent source-to-binary reconstruction of the historical executable. The post-freeze audit defines a concrete release procedure: begin

the next experiment from a committed source revision and an attested build, rerun the public 1.0.6 checks on held-out tasks, and publish the complete package described above.

References

- [1] Green Software Foundation. Software carbon intensity specification, version 1.1.0. Technical report, Green Software Foundation, 2024.
- [2] Pengfei Li, Jianyi Yang, Mohammad A. Islam, and Shaolei Ren. Making AI less “thirsty”: Uncovering and addressing the secret water footprint of AI models. *Communications of the ACM*, 68(7):54–61, 2025.
- [3] George C. Necula. Proof-carrying code. In *Proceedings of the 24th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages*, pages 106–119, 1997.
- [4] Jingxi Qiu, Zeyu Han, and Cheng Huang. SURE-RAG: Sufficiency and uncertainty-aware evidence verification for selective retrieval-augmented generation, 2026.
- [5] Dongyu Ru, Lin Qiu, Xiangkun Hu, Tianhang Zhang, Peng Shi, Shuaichen Chang, Chengwei Wang, Shichao Sun, Huanyu Li, Zizhao Pandey, Sirui Wang, Tuan Dung Nguyen, Edward Y. Ma, Qian Peng, Zhengzhong Zeng, Chun-Hao Yu, Hanieh Deilamsalehy Chawla, Zhenchang Xu, Bryan Hsieh, Sunil Nayak, Timmy Zhang, Han Wang, Caiming Cheng, Yangqiu Yin, and Heng Ji. RAGChecker: A fine-grained framework for diagnosing retrieval-augmented generation, 2024.
- [6] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948.
- [7] Arman Shehabi, Sarah J. Smith, Alex Hubbard, Alex Newkirk, Nuoa Lei, Md Abu Bakar Siddik, Billie Holecek, Jonathan Koomey, Eric Masanet, and Dale Sartor. 2024 united states data center energy usage report. Technical Report LBNL-2001637, Lawrence Berkeley National Laboratory, 2024.
- [8] Yujie Zhao, Boqin Yuan, Junbo Huang, Haocheng Yuan, Zhongming Yu, Haozhou Xu, Lanxiang Hu, Abhilash Shankarampeta, Zimeng Huang, Wentao Ni, Yuandong Tian, and Jishen Zhao. AMA-Bench: Evaluating long-horizon memory for agentic applications, 2026.